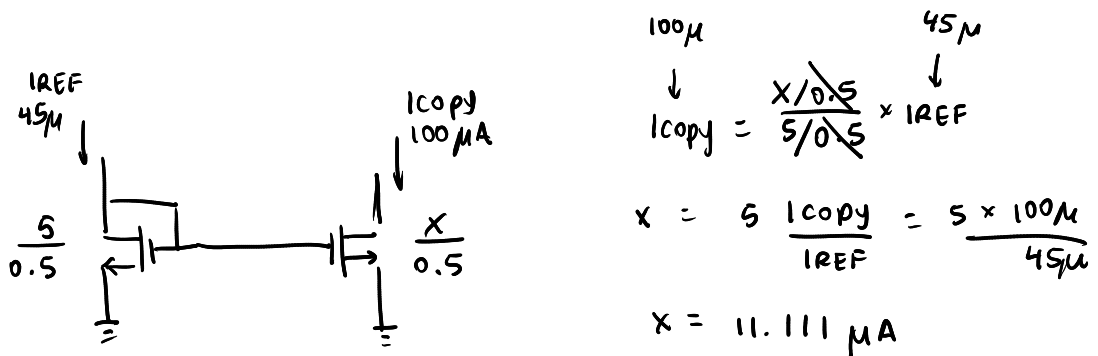




$$g_{m4} = \sqrt{2 \kappa' \frac{W}{L} I_D} = \sqrt{2 \mu_n C_{ox} \frac{W}{L} I_D}$$

$$8.85 \times 10^{-12} \frac{F}{m} \times \frac{1m}{100cm}$$

$$C_{ox} = \frac{\epsilon_0 \cdot \epsilon_r}{t_{ox}} = \frac{8.85 \times 10^{-14} \text{ F/cm} \times 3.9}{1.39 \times 10^{-8} \text{ m} \times \frac{100 \text{ cm}}{1 \text{ m}}} = 248.4 \text{ nF/cm}^2$$

$$g_{m4} = \sqrt{2 \left(458.4 \frac{\text{cm}^2}{\text{Vs}} \right) \left(248.4 \frac{\text{nF}}{\text{cm}^2} \right) \left(\frac{5}{6.5} \right) (22.5 \mu\text{A})} = 226.4 \mu\text{S}$$

$$g_{m7} = \sqrt{2 \left(212.0 \frac{\text{cm}^2}{\text{Vs}} \right) \left(248.4 \frac{\text{nF}}{\text{cm}^2} \right) \left(\frac{19}{0.5} \right) (75 \mu\text{A})} =$$

$$r_0 = \frac{1}{210}$$